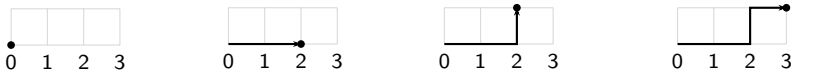
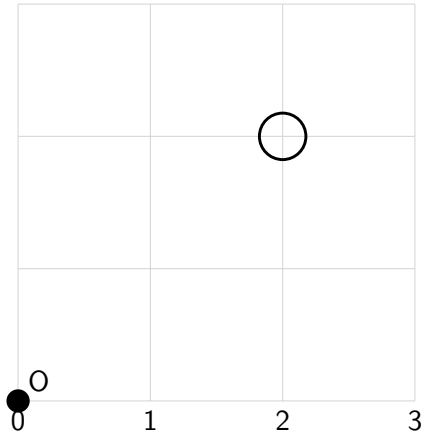
Move one grid edge per card: E $\rightarrow$, W $\leftarrow$, N $\uparrow$, S $\downarrow$. For N, add the column number to the memory. For S, subtract it. E and W leave the memory unchanged.

One partner moves the robot; the other changes its memory marker and checks each move. Use the loose memory strip from page 5 for every problem. Keep the cards in order, or write a longer route as letters. Each trial starts at the marked dot with memory 0.



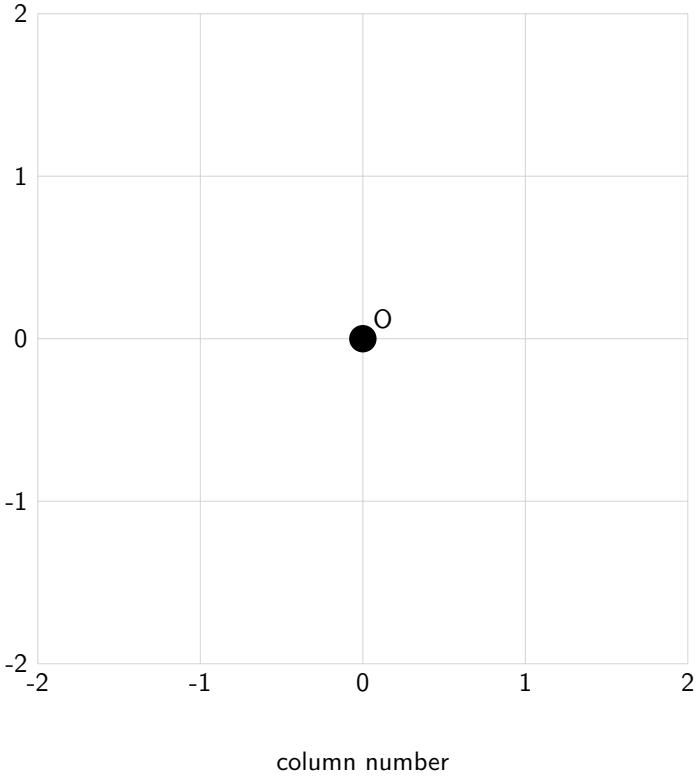
start: memory 0 EE: memory 0 EEN: memory 2 EENE: memory 2

Problem 1: Use two E cards and two N cards, in any order. Find every different route from O to the ring. What memories can you leave there?



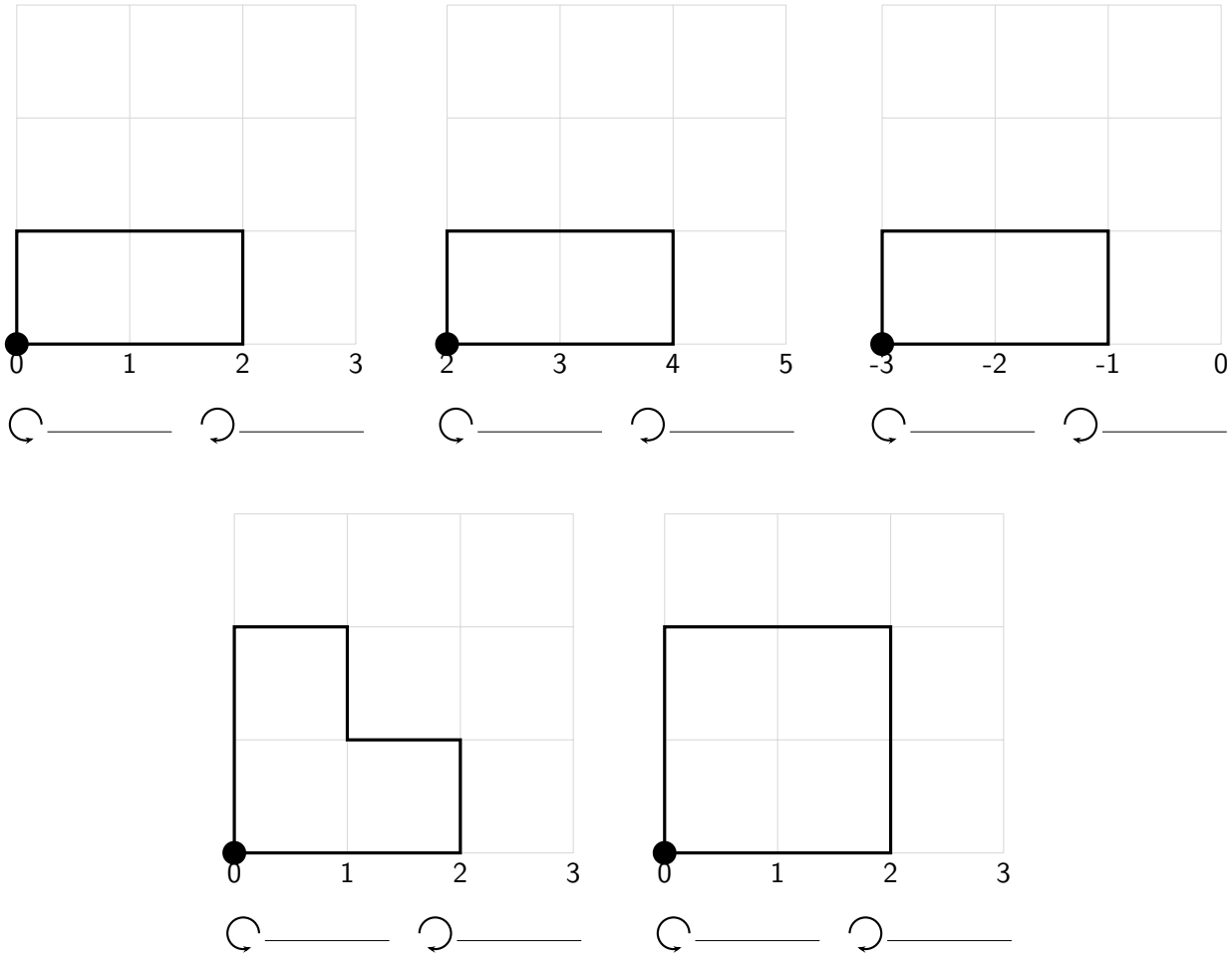
route final memory

Problem 2: Use E, N, W and S once each. Which memories can you leave when you return to O? Keep one route for each possible memory. Can any other memory occur with these four cards?

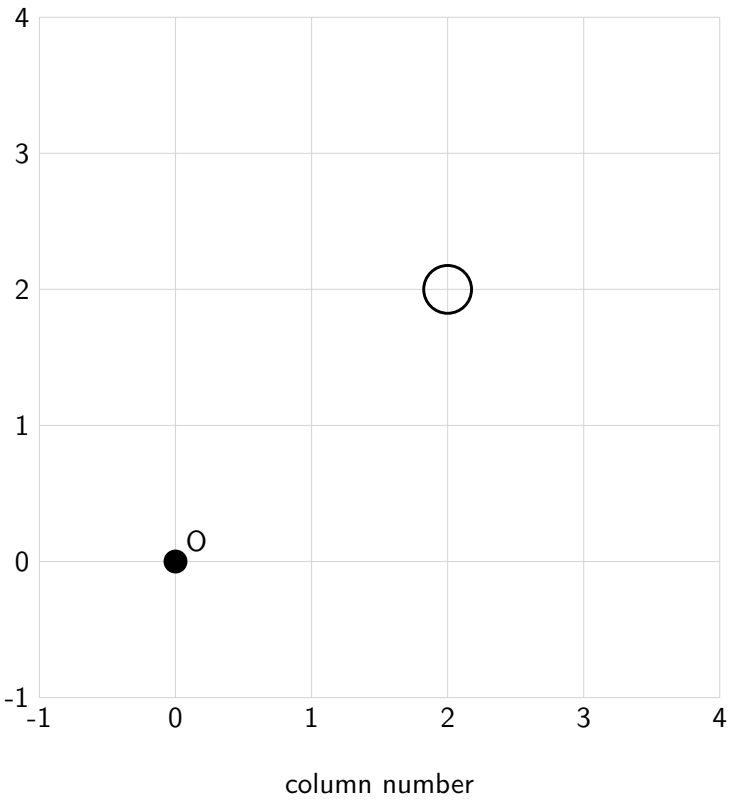


route | final memory

Problem 3: Start at each dot with memory 0. Walk once around its outline in each direction. How does moving the same rectangle to a new place affect its final memory? What connects a loop's shape, direction and final memory?



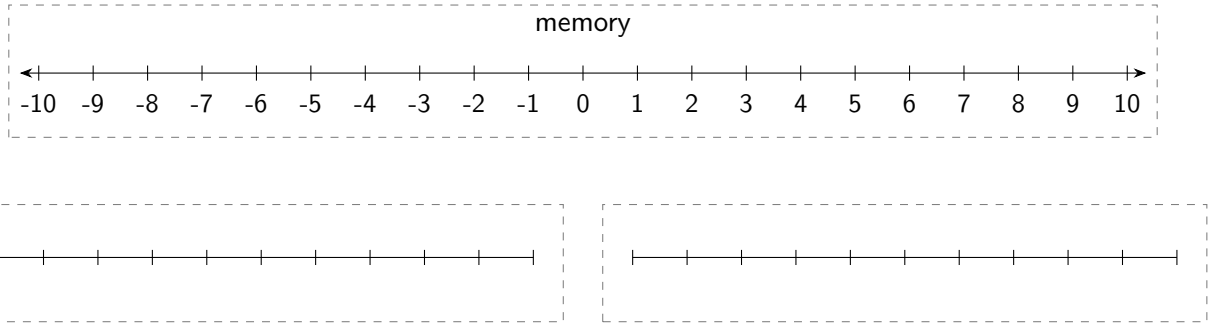
Problem 4: Start at O with memory 0. Reach the ring with memory 7. Then start again and reach it with memory -3. You may use any number of E, N, W and S moves. Can you reach the ring with any integer memory (positive, zero or negative)? Explain a plan.



route to memory 7: _____

route to memory -3: _____

Cut out the memory strip and keep it beside your working page. If you need more numbers, tape on an extension, overlapping one tick, and continue numbering. Make more extensions with the same tick spacing when needed.



Cut out the move cards. For a route longer than your cards allow, write its letters in order on paper.

E →	E →	E →	E →	E →	E →	E →	E →
W ←	W ←	W ←	W ←	W ←	W ←	W ←	W ←
N ↑	N ↑	N ↑	N ↑	N ↑	N ↑	N ↑	N ↑
S ↓	S ↓	S ↓	S ↓	S ↓	S ↓	S ↓	S ↓